



Problem solving with non-linear relationships

- 1 This table shows the value of a car two years after it was made. If the values form a geometric sequence, what is the common multiplier between terms? (You may use your calculator.) Tick **1** correct answer

Year	Value of car (£)
0	20 000
1	16 000
2	12 800
3	

- ☐ 0.2
☐ 0.4
☐ 0.6
☒ 0.8
☐ 1.25

- 2 This table shows the value of a car 2 years after it was made. If the values continue to follow the same geometric sequence, what is the car's value after 3 years? (You may use your calculator.) £ 10 240 Fill in the blank

Year	Value of car (£)
0	20 000
1	16 000
2	12 800
3	



- 3 This table shows the value of a car two years after it was made. Why might this sequence not continue as the same geometric sequence forever? Tick **2** correct answers

Year	Value of car (£)
0	20 000
1	16 000
2	12 800
3	

- ☐ If the geometric sequence continues the value will drop below zero pounds.
- ☐ Sequences cannot go on forever, they will eventually stop.
- ☒ The value of the car may stop decreasing, it could reach minimum value.
- ☒ The value may decrease sharply in the first 2 years, then the pattern may change
- ☐ The car loses the most value in the first year, not the same amount each year.

- 4 Which two sequences can be added to make the first 5 terms of a linear sequence? Tick **2** correct answers

- ☒ 3, 7, 11, 15, 19, ...
- ☐ 5, 10, 20, 40, 80, ...
- ☐ 6, 7, 9, 12, 16, ...
- ☒ 14, 11, 8, 5, 2, ..
- ☐ 16, 24, 36, 54, 81, ...



- 5 Order these terms so they form part of a Collatz sequence: "If the term is even, the next term is half the previous term. If the term is odd, the next term is 3 times the previous term add 1". Use numbers to show the correct order

2	100
4	25
1	33
5	76
6	38
3	50

- 6 Which two sequences can be added to make the first 5 terms of a linear sequence? Tick 2 correct answers

- ☒ 1, 3, 8, 16, 27, ...
- ☐ 8, 11, 13, 14, 14, ...
- ☒ 10, 19, 25, 28, 28, ...
- ☐ 15, 16, 19, 24, 31, ...
- ☐ 20, 17, 14, 11, 8, ...

